Zen AI Model Family

Zen-Designer-235B-A22B-Instruct

Vision-Language Design Assistant

Technical Whitepaper v1.0

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Abstract

We present **Zen-Designer-235B-A22B-Instruct**, an instruction-tuned vision-language model for design and visual-creation workflows. The model is a derivative of the open-weight **Qwen3-VL-235B-A22B** mixture-of-experts foundation (Apache 2.0), adapted by Hanzo AI and the Zoo Labs Foundation for UI/UX analysis, layout reasoning, and visual question answering. It is a sparse MoE with 235B total parameters of which approximately 22B are active per token, served under the `Qwen3VLForConditionalGeneration` architecture. Weights are released openly under Apache 2.0 at huggingface.co/zenlm/zen-designer-235b-a22b-instruct. This whitepaper documents the architecture as published in the released configuration and the intended use of the instruction-tuned variant; it does not claim novel pretraining or report benchmark numbers that have not been independently reproduced.

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1 Introduction

Zen-Designer-235B-A22B-Instruct is the instruction-tuned member of the Zen-Designer line, intended for direct (non-deliberative) design and visual-analysis tasks. It is built on the open-weight **Qwen3-VL-235B-A22B** foundation released by the Qwen team under the Apache 2.0 license, and is distributed as part of the Zen AI Model Family. A companion *thinking* variant (zen-designer-235b-a22b-thinking) trades latency for extended deliberation.

1.1 Provenance and Licensing

This model is a derivative of an upstream open-weight foundation. We state the lineage explicitly to avoid any overclaiming:

- **Upstream foundation**: Qwen3-VL-235B-A22B (vision-language MoE), Apache 2.0.
- **Derivative**: Zen-Designer-235B-A22B-Instruct, instruction-tuned for design tasks.
- **License**: Apache 2.0 (inherited; redistribution permitted with attribution).
- **Repository**: `zenlm/zen-designer-235b-a22b-instruct` on Hugging Face.

1.2 Scope

- **Sparse MoE efficiency**: ∼22B parameters active per token from a 235B total pool.
- **Instruction tuning**: optimized for direct execution without a separate thinking phase.
- **Vision-language**: image-conditioned analysis and text-conditioned design assistance.
- **Extended context**: 131K-token maximum position embeddings (per released config).

2 Architecture

2.1 Model Configuration

The values below are taken directly from the released model configuration (`config.json`); they describe the published architecture rather than any internally measured performance.

Component	Specification
Architecture	`Qwen3VLForConditionalGeneration`
Total Parameters	235B (sparse MoE)
Active Parameters	∼22B per token
Experts / Active per Token	64 / 4
Text Hidden Size	8192
Text Layers	80
Attention Heads / KV Heads	64 / 8
Vision Hidden Size	2048
Vision Layers	48
Vision Patch / Image Size	14 / 2048
Vocabulary Size	151,936
Max Position Embeddings	131,072
Precision	bfloat16

Table 1: Zen-Designer-235B-A22B-Instruct configuration (from released `config.json`).

2.2 Mixture of Experts

The text backbone is a sparsely activated mixture-of-experts transformer: each token is routed to a small subset (top-4 of 64) of expert feed-forward blocks per layer, so that only a fraction of the 235B total parameters is exercised on any given forward pass. This is the standard MoE efficiency trade-off described by Shazeer et al. [3] and Fedus et al. [4], inherited from the Qwen3-VL foundation.

2.3 Vision-Language Integration

A vision encoder (48 layers, hidden size 2048, patch size 14) produces visual tokens that are consumed by the text decoder under the `Qwen3VLForConditionalGeneration` interface, enabling image-conditioned design analysis alongside text-only generation.

3 Intended Use

3.1 Primary Applications

- UI/UX design analysis and critique
- Layout and component-structure reasoning
- Visual question answering over screenshots and mockups
- Design-system documentation assistance
- Accessibility review of interfaces

3.2 Instruct vs. Thinking

The instruct variant is tuned for low-latency, direct responses. Where multi-step visual reasoning or self-verification is required, the companion thinking variant is preferred at the cost of additional latency.

3.3 Usage Example

```
from transformers import AutoModelForVision2Seq, AutoProcessor

model = AutoModelForVision2Seq.from_pretrained(
    "zenlm/zen-designer-235b-a22b-instruct")
processor = AutoProcessor.from_pretrained(
    "zenlm/zen-designer-235b-a22b-instruct")

# Image-conditioned analysis
inputs = processor(images=image, text="Analyze this UI", return_tensors
   ="pt")
outputs = model.generate(**inputs)
analysis = processor.decode(outputs[0])
```

Listing 1: Loading and inference

4 Deployment

4.1 Available Formats

- **SafeTensors**: native bfloat16 weights (sharded, ∼96 shards).
- **GGUF / MLX**: community quantizations may be provided separately.

4.2 Hardware Footprint

Because the model carries 235B total parameters, full-precision serving requires multi-accelerator deployment; quantization reduces this footprint at some quality cost. We do not publish a single canonical latency figure, as throughput depends heavily on the serving stack, batch size, quantization, and hardware.

5 Limitations

- **Inherited behavior**: as a derivative of Qwen3-VL, the model inherits the capabilities, biases, and limitations of its upstream foundation and training data.
- **No independent benchmark claims**: this document intentionally omits benchmark scores that have not been independently reproduced for this derivative.
- **Resource intensity**: the 235B parameter count makes self-hosting costly relative to smaller members of the Zen family.
- **Visual generation**: the model assists with design *reasoning* and analysis; pixel-level image synthesis is out of scope for the language backbone.

6 Conclusion

Zen-Designer-235B-A22B-Instruct packages an open-weight Qwen3-VL-235B-A22B foundation as an instruction-tuned design and visual-analysis assistant, released openly under Apache 2.0. We have described its architecture as published and stated its provenance plainly so that downstream users can make informed decisions about licensing, capability, and cost.

Acknowledgments

We thank the Qwen team for releasing the Qwen3-VL foundation under Apache 2.0, and the open-source community and the teams at Hanzo AI and Zoo Labs Foundation for their contributions to this work.

References

[1] Vaswani, A. et al. (2017). Attention Is All You Need. NeurIPS 2017.

[2] Radford, A. et al. (2021). Learning Transferable Visual Models From Natural Language Supervision. ICML 2021.

[3] Shazeer, N. et al. (2017). Outrageously Large Neural Networks: The Sparsely-Gated Mixture-of-Experts Layer. ICLR 2017.

[4] Fedus, W. et al. (2022). Switch Transformers: Scaling to Trillion Parameter Models with Simple and Efficient Sparsity. JMLR 2022.

[5] Ouyang, L. et al. (2022). Training Language Models to Follow Instructions with Human Feedback. NeurIPS 2022.

[6] Qwen Team (2025). Qwen3-VL: Vision-Language Models. huggingface.co/Qwen.

A Model Card

Field	Value
Model Name	Zen-Designer-235B-A22B-Instruct
Upstream Foundation	Qwen3-VL-235B-A22B (Apache 2.0)
Version	1.0.0
Release Date	September 2025
License	Apache 2.0
Repository	huggingface.co/zenlm/zen-designer-235b-a22b-instruct
Documentation	github.com/zenlm/zen
Contact	research@hanzo.ai

Table 2: Model Card Information